SM2501/FSG3084 Biomechanics of human movement
HT25

**Project 2**

**Video recording of project presentation**

For this assignment, you will create a video recording of the project presentation. A separate presentation template and tips for recording is provided. The presentation should be *no more than 15 mins*. All group members need actively participate in the presentation. The video recording and a PDF version of the presentation need to be uploaded to your group's project page in Canvas.

***Due: Nov 27th, 12:00***. The feedback of this assignment will be given orally. Each group will be asked to book a 10 mins ZOOM meeting. Detailed information about the oral feedback session will come later!

We have uploaded marker data (actual markers, not computed joint centers) and ground reaction force data (static and dynamic marker data are stored in .trc files and ground reaction force data is stored in .mot file). An example input setting file for walking has also been uploaded. For each motion, we have also provided a file with measured EMG activities. The EMG recording includes the rectus femoris, vastus lateralis, tibialis anterior, gastrocnemius medialis, soleus, and bicep femoris long head. The EMG data was post-processed and has a scale of 0 to 1, wherein 1 indicates full activation and 0 indicates no activation. You are to accomplish the following tasks:

**Motions to analyze:**

**ALL groups**

- Normal walking (*Left Side Only*), which has valid force plate data during the entire gait cycle. Time interval 5.11s – 6.09s
- Crouch gait (*Left Side Only*), i.e., time interval: 2.76s – 3.75s
- Jogging (*Left Side Only*), i.e., time interval: 1.66s – 2.36s

Screenshots of movements are attached.

**TO DO**

**Part 1**

**Scale the generic model, then perform inverse kinematics (IK) and inverse dynamics (ID) on each motion in OpenSIM.**

- Plot ankle, knee and hip joint moment (All three planes in the hip, sagittal plane in the knee and ankle). Make a separate plot for each joint and plane. Discuss your plots qualitatively and quantitively. *Tips:* take ground reaction force vector into consideration when analyzing the underlying mechanism.

- Compare joint moment plots computed in OpenSim with the reference plots computed by Vicon (hip, knee and ankle in sagittal plane). Are the joint moments computed by two different tools comparable? If not, discuss the possible reason.

**Part 2**
**Perform residual reduction (RRA) and static optimization (SO) on each motion**

**For all three motions**

- Plot **activations** vs. **time** for hip extensors, hip flexors, hip abductors, knee extensors and ankle plantarflexors and dorsiflexors (one muscle group per figure). Compare your findings with measured EMG activity for available muscles (supplied in the ExperimentalData, plot the measured EMG and computed muscle activation in the same figure). Observe and discuss similarities and/or differences.
- Discuss which major muscles contribute to each motion (including how the muscles contract concentrically, eccentrically, or isometrically)
- Compare the muscle activation between ***Crouch Gait*** and ***Normal Walking***, discuss similarity and differences.
- Compare the muscle activation between ***Jogging*** and ***Normal Walking***, discuss similarity and differences.
- Find a time interval with significant joint co-contraction (i.e., opposing/antagonistic muscle groups, for instance, both extensors and flexors, are both activated at the same time at the same joint) in either ***Crouch Gait*** or ***Jogging***. Discuss and rationalize this co-contraction.

**Part 3**
**Methodology discussion**

- What is the purpose of performing RRA?
- What are the differences between residual actuators and reserve actuators?
- Are you satisfied with the RRA results? State the reason. Please also comment on the recommended mass changes after the RRA in different motions.
- What are the ground assumptions and limitations in SO?

For all motions, appendix **force plot for reserve actuators** (plot all **reserve actuators** in one figure, *not residual actuator*)

**IMPORTANT INFORMATION**:

1. The example setting files were prepared based on the Windows system. We recommend that you perform simulations on a Windows computer if possible.
2. *You are to modify example setting files* (.xml) when appropriate to fit your own simulation and different motions, e.g. mass, height, time range, file path (If you open setting files, some file path in the example setting files are based on folders from my computer) and names, weighting factors (usually in the setting files with the name _task.xml) and optimal force for the actuators

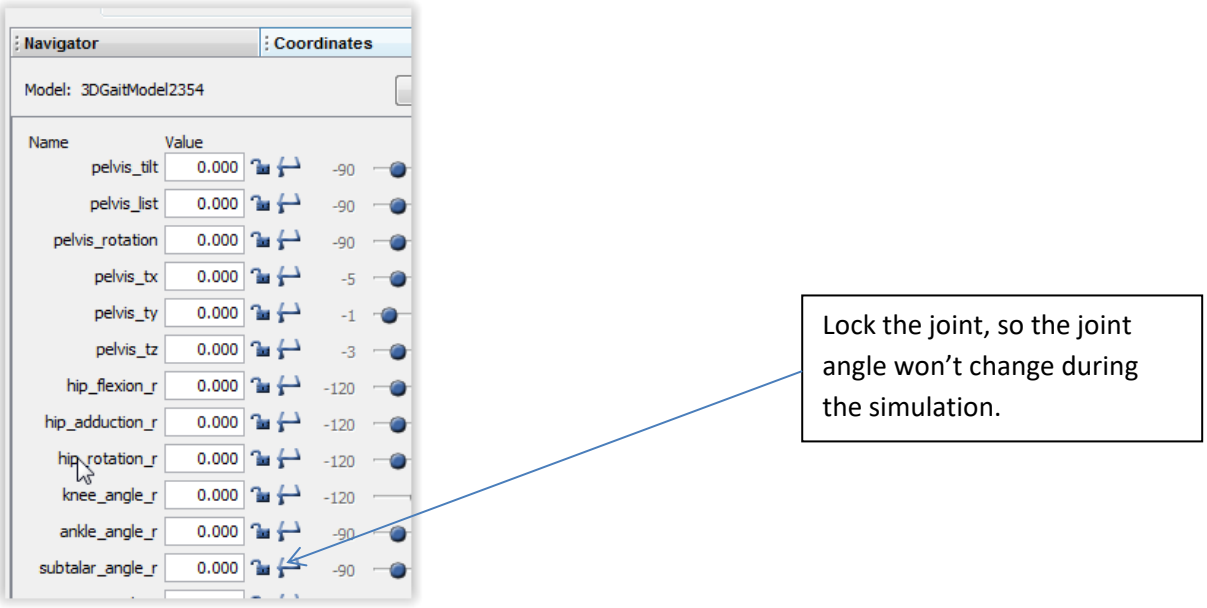
(in RRA and Static Optimization) etc. Detailed introductions about input setting files can be found in the **OpenSim User's Guide**.

3. Make sure you change the application location of the point residual actuators to the **mass center of pelvic** of your scaled model in actuators files for RRA and CMC (Appendix A).
4. In RRA, the center of mass adjustment should be applied on **Torso** segment. The recommended changes in each segment's mass are *not* automatically applied in the RRA-adjusted model – a manual modification is needed.
5. The subject's height is 1670 mm and weight is 70.7 kg.
6. For simplification, do not forget to **LOCK** the **subtalar joint** and the **mtp joint** after scaling (before performing any further analysis of IK, ID, RRA, and SO, Appendix B).
7. You can review the marker fitness in Scaling and IK by previewing experimental marker data and overlay experimental markers on top of the virtual markers (Appendix C).
8. If there are markers missing from the .trc file (due to the low quality of the marker data), you may experience error in IK. The easiest solution is to uncheck the 'enable box' for the missing marker.
9. In case you experience error in RRA or very slow convergence in performing Static Optimization: For RRA, you can try with 2-3 iterations in order to achieve a desired residual force and moment. For static optimization, you can choose an appropriate 'maximum_number_of_integrator_steps' and 'output_precision' in the setup file.
10. Turn reserve and residual actuators carefully to ensure a physically meaningful computational result in Static Optimization.
11. For **jogging and/or crouch**, you can increase the weight of RRA tracking task for pelvis ty axis (default value is 5) to 15 or 20. To do this, you can open file gait2392_RRA_Tasks.xml, and change the weight value at line 208 (Appendix D).
12. **About Plot**:
    a. Prepare your plot in any programming language you like (e.g., Matlab). Make sure you have axes labels (indicating the sign convention in the axes labels), units, and legends. An example matlab script for plotting is provided.
    b. Follow the same sign convention as the first project (pay attention that OpenSim has an opposite sign convention on knee flexion and extension)
    c. Plot only the time interval specified for each movement.
    d. Make sure the **Font Size** and **Line Thickness** can be clearly viewed on screen.

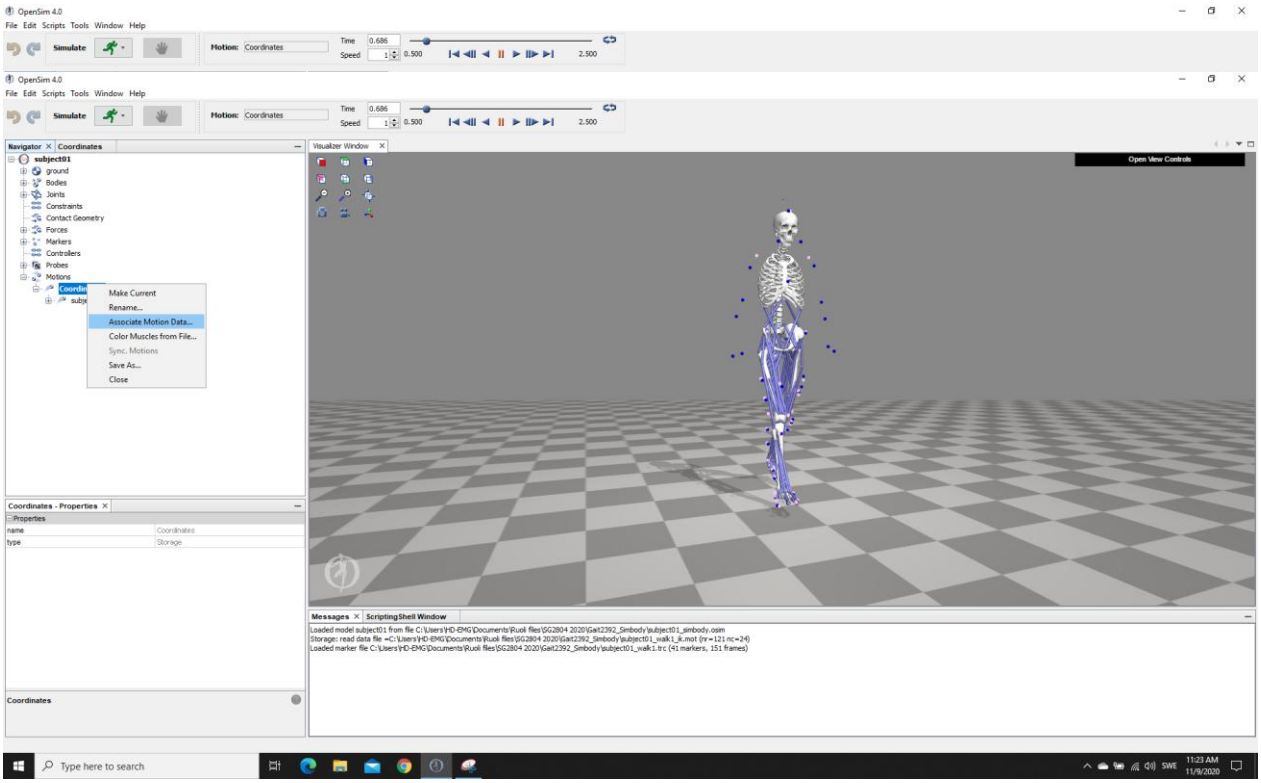
## Appendix A

```
  <PointActuator name="FX">
            <isDisabled> false </isDisabled>
            <!--Minimum allowed value for control signal. Used primarily when solving
                for control values-->
            <min_control> -infinity </min_control>
            <!--Maximum allowed value for control signal. Used primarily when solving
                for control values-->
            <max_control> infinity </max_control>
      <body> pelvis </body>
            <point>      -0.0743679       0.00000000000000000000       0.00000000000000000000 </point>
      <point_is_global> false </point_is_global>
            <direction>       1.00000000000000000000      -0.00000000000000000000      -0.00000000000000000000 </dire
      <force_is_global> true </force_is_global>
            <optimal_force>       4.00000000000000000000 </optimal_force>
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            <!--Minimum allowed value for control signal. Used primarily when solving
                for control values-->
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            <!--Maximum allowed value for control signal. Used primarily when solving
                for control values-->
            <max_control> infinity </max_control>
  <body> pelvis </body>
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                for control values-->
            <min_control> -infinity </min_control>
            <!--Maximum allowed value for control signal. Used primarily when solving
                for control values-->
            <max_control> infinity </max_control>
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            <point>      -0.0743679       0.00000000000000000000       0.00000000000000000000 </point>
```

## Appendix B



## Appendix C

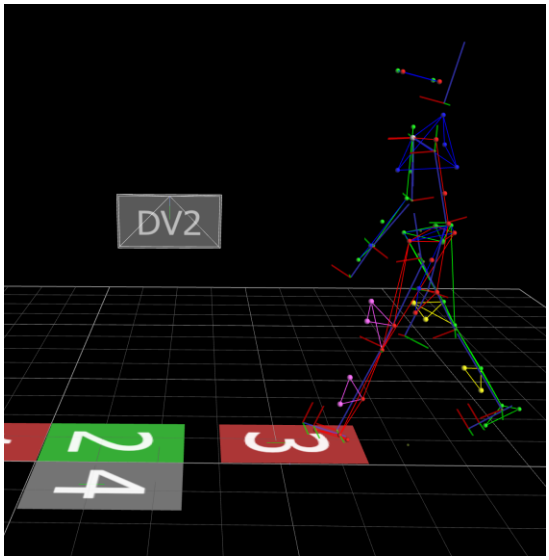


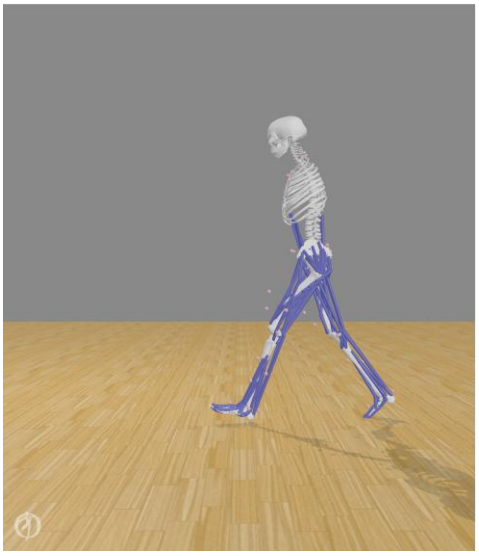

## Appendix D

```
        </CMC_Joint>
        <CMC_Joint name="pelvis_tx">
            <!--Flag (true or false) indicating whether or not a task is enabled.-->
            <on>true</on>
            <!--Weight with which a task is tracked relative to other tasks. To track a task more tightly, make the weight larger.-->
            <weight> 5</weight>
            <!--Name of body frame with respect to which a tracking objective is specified. The special name 'center_of_mass' refers to the system center of mass. This property
            <wrt_body>-1</wrt_body>
            <!--Name of body frame in which the tracking objectives are expressed. This property is not used for tracking joint angles.-->
            <express_body>-1</express_body>
            <!--Array of 3 flags (each true or false) specifying whether a component of a task is active. For example, tracking the trajectory of a point in space could have t
            <active>true false false </active>
            <!--Position error feedback gain (stiffness). To achieve critical damping of errors, choose kv = 2*sqrt(kp).-->
            <kp> 100 1 1</kp>
            <!--Velocity error feedback gain (damping). To achieve critical damping of errors, choose kv = 2*sqrt(kp).-->
            <kv> 20 1 1</kv>
            <!--Feedforward acceleration gain. This is normally set to 1.0, so no gain.-->
            <ka> 1 1 1</ka>
            <!--Direction vector[3] for component 0 of a task. Joint tasks do not use this property.-->
            <r0> 0 0 0</r0>
            <!--Direction vector[3] for component 1 of a task. Joint tasks do not use this property.-->
            <r1> 0 0 0</r1>
            <!--Direction vector[3] for component 2 of a task. Joint tasks do not use this property.-->
            <r2> 0 0 0</r2>
            <!--Name of the coordinate to be tracked.-->
            <coordinate>pelvis_tx</coordinate>
            <!--Error limit on the tracking accuracy for this coordinate. If the tracking errors approach this limit, the weighting for this coordinate is increased. -->
            <limit>0</limit>
        </CMC_Joint>
        <CMC_Joint name="pelvis_ty">
            <!--Flag (true or false) indicating whether or not a task is enabled.-->
            <on>true</on>
            <!--Weight with which a task is tracked relative to other tasks. To track a task more tightly, make the weight larger.-->
            <weight> 15</weight>
            <!--Name of body frame with respect to which a tracking objective is specified. The special name 'center_of_mass' refers to the system center of mass. This property
            <wrt_body>-1</wrt_body>
            <!--Name of body frame in which the tracking objectives are expressed. This property is not used for tracking joint angles.-->
            <express_body>-1</express_body>
            <!--Array of 3 flags (each true or false) specifying whether a component of a task is active. For example, tracking the trajectory of a point in space could have t
            <active>true false false </active>
            <!--Position error feedback gain (stiffness). To achieve critical damping of errors, choose kv = 2*sqrt(kp).-->
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            <!--Velocity error feedback gain (damping). To achieve critical damping of errors, choose kv = 2*sqrt(kp).-->
            <kv> 20 1 1</kv>
            <!--Feedforward acceleration gain. This is normally set to 1.0, so no gain.-->
            <ka> 1 1 1</ka>
            <!--Direction vector[3] for component 0 of a task. Joint tasks do not use this property.-->
            <r0> 0 0 0</r0>
```

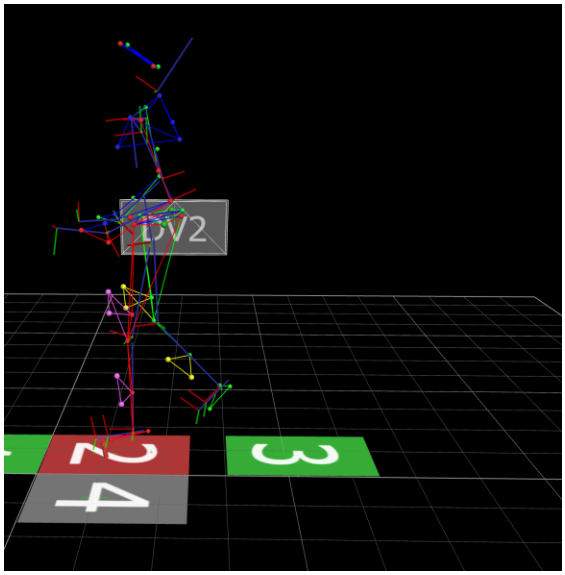
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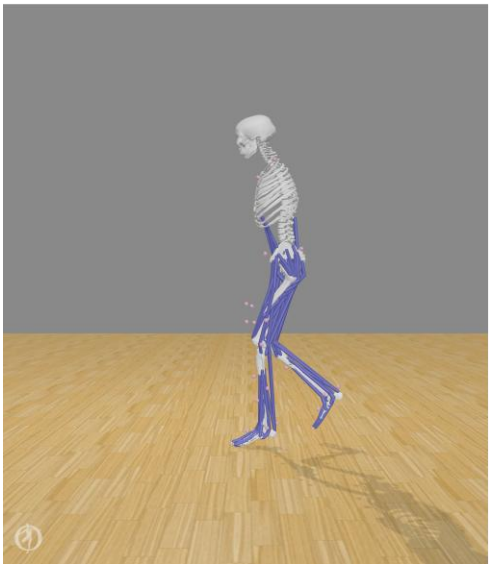
Normal Gait



Jogging



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HT25

Crouch Gait

